



Assignment: Exploratory Data Analysis

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GMI35S

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The Architectural Design of Animal Crossing Villagers

1. Introduction & Dataset Overview

This report explores the structure behind the character system in *Animal Crossing: New Horizons*. The dataset consists of 391 unique villagers, each described by 17 attributes, resulting in over 6,000 data points. This provides a solid foundation for exploratory data analysis (EDA).

The goal of this analysis was to determine whether villager traits are randomly distributed or if there is an underlying design structure. Most variables in the dataset are categorical, such as Species, Personality, Hobby, and Gender.

Data Quality Assurance

Before starting the analysis, the dataset was checked for missing values and duplicates. No issues were found, meaning the data was complete and suitable for analysis.

2. Core Insights & Findings

2.1 Species and Personality Distribution

The initial analysis revealed that species are not evenly distributed. While there are 35 different species, some appear much more frequently than others. For example, cats are significantly more common, which may reflect design choices focused on player familiarity and preference.

In contrast, personality types are more evenly distributed. However, certain types such as *Lazy* and *Normal* appear slightly more often. This aligns with the game's relaxed and low-stress design, where calm personalities support the overall atmosphere.

2.2 Birth Month Distribution

The "Birthday" variable was converted into a numerical "Birth Month" feature. The results show that birthdays are evenly distributed throughout the year, with no strong clustering in specific months.

This suggests that an intended design choice was made to ensure that game events, such as villager birthdays, happen consistently over time maintaining player engagement.

2.3 Relationships Between Traits

To examine relationships between variables, categorical data was transformed using one-hot encoding and analyzed using a correlation heatmap.

The results show that most relationships are weak, but some clear patterns exist.

For example:

- A strong relationship between *Jock* personality and *Fitness* hobby
- A noticeable connection between *Peppy* or *Snooty* personalities and *Fashion*

These patterns suggest that certain character traits are intentionally linked.

2.4 Personality and Birth Month

A boxplot analysis was used to compare birth month across different personality types.

The results show similar distributions for all personalities, with no strong seasonal patterns. This indicates that personality is independent of birth timing and that birthdays are distributed evenly across all character types.

3. Predictive Analysis

Since the dataset mainly contains categorical variables, traditional regression analysis was not suitable. Instead, a classification approach was used.

A decision tree model was implemented to predict a villager's hobby based on their personality.

The model achieved an accuracy of 0.51. Given that there are six possible hobbies, a random guess would result in an accuracy of approximately 16.7%. This means the model performs significantly better than random prediction.

This result confirms that personality provides useful information for predicting behavior, supporting the earlier findings that certain traits are structured and related.

4. Conclusion

This analysis shows that the world of *Animal Crossing* is built on a structured and well-designed system rather than randomness.

Key observations include:

1. Uneven distribution of species, suggesting intentional variation
2. Even distribution of birthdays, supporting consistent gameplay
3. Clear relationships between personality and hobby
4. Predictable patterns that can be used for classification

Overall, the dataset demonstrates how the developers combine structure and variation to create a balanced and engaging player experience.